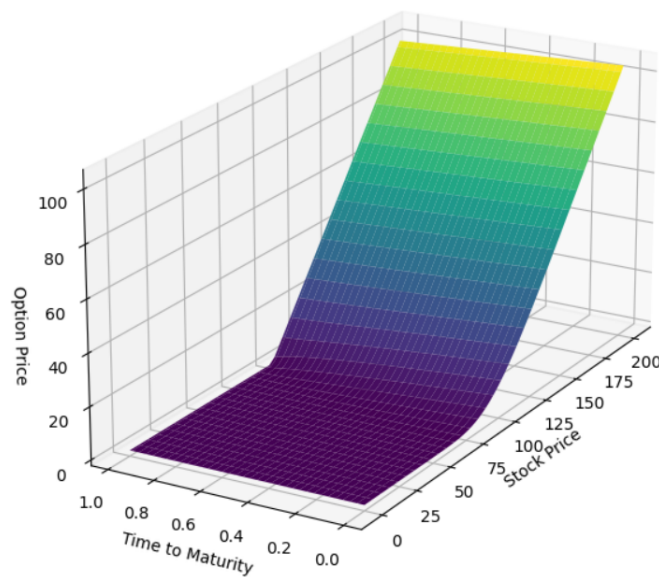


The Black–Scholes Equation: Transforming Quantitative Finance by Hedging Risk and Catalyzing New Industries

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1 Introduction

Around the year 600 BC, ancient Greek mathematician and philosopher Thales of Miletus predicted that the season's olive harvest would be unusually abundant, which would increase the demand for olive presses. Although he could not afford to buy his own olive presses, Thales paid a smaller amount of money to acquire the right to rent the olive presses for a specified price during the harvest season. When the harvest came, there were indeed more olives than expected, so Thales made a profit by renting out the presses at a much higher price than he paid.

Thales was the first reputed buyer of a *call option*, the right but not the obligation to buy something at a later date for a pre-specified price (the strike price). Similarly, a *put option* is the right but not the obligation to sell something at a later date for a pre-specified price (the strike price). So, if one predicts that a stock will increase in value, as Thales did, they would benefit from buying a call option, and if they predict that a stock will decrease in value, they would benefit from buying a put option.

There are several important advantages of options. First, they can often provide higher returns relative to the initial cost. Second, they can offer lower risk (compared to equities) through hedging. Third, options can provide leveraging power through stock replacement. Fourth, options can offer the flexibility needed for strategies such as spreads, straddles, and strangles. Quantitative investors can take advantage of these benefits with a thorough understanding of the mathematics underpinning the use of options in financial economics.

2 Determining the value of an option

The question that all economic markets seek to answer is, “How much should this be worth?” The most common method of determining such a value is summing its future cashflows, discounted to present value. However, the future cashflows of an option are uncertain, because they depend on the future stock price. So, we can attempt to find a trading strategy that will have the same payout as the option, and then set the price of the option equal to the price of the strategy.

Such a trading strategy would need to have the exact same risks as the option. Thus, we can create a portfolio with both the option and a certain number of shares (involved in the trading strategy) in which the risks would be offset, meaning that the portfolio is risk-free.

We denote the price of the option as $V(s, t)$, where s is the stock price and t is time. Moreover, we denote the number of shares as Δ and the share price as S . Then, the portfolio Π is given by $\Pi = V(s, t) - \Delta S$. To analyze the risk factors, we need to look at the evolution of the portfolio, specifically the change in the portfolio value. We have $d\Pi = dV - \Delta dS$, where the d refers to an infinitesimal change in the variable that follows it.

2.1 Change in the stock price

First, let us consider the change in the stock price. Initially, it was believed that since the stock price is dependent on so many variables, it is virtually impossible to predict all of them accurately. So, the best we can do is assume that at any given point in time, the stock price is equally likely to increase or decrease. In other words, the stock price follows a random walk, which can be modeled with **Brownian motion**.

However, this is not accurate, as a stock price often increases over time if the business is doing well or decreases over time if the business is doing poorly. Thus, to account for this drift we choose to model stock prices with **geometric Brownian motion**, which satisfies the equation

$$dS = \mu S dt + \sigma S dW.$$

Here, μ (the percentage drift) and σ (the percentage volatility) are constants, and W represents Brownian motion. Hence, $\mu S dt$ represents the drift, or growth rate, of the stock price, and $\sigma S dW$ represents the volatility of the stock price.

2.2 Change in option price

By Itô's Lemma,

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \cdot \frac{\partial^2 V}{\partial S^2} dS^2.$$

Substituting our equation for dS into the equation above and noting that in stochastic calculus, $dt \cdot dt = dt \cdot dW = dW \cdot dW = 0$, we obtain

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 + \frac{\partial^2 V}{\partial S^2} dt.$$

2.3 The Black-Scholes equation

Substituting the equation above into $d\Pi = dV - \Delta dS$ and factoring out dt and dS , we get

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \left(\frac{\partial V}{\partial S} - \Delta \right) dS.$$

Now, the second term represents the volatility, or risk. Since we established that Π is risk-free, we set $\frac{\partial V}{\partial S} - \Delta = 0$, which yields $\Delta = \frac{\partial V}{\partial S}$.

However, a risk-free portfolio should yield a risk-free rate, so $d\Pi = r \cdot \Pi dt$, where r is the risk-free interest rate. Substituting the expression for Π into this equation, we get

$$d\Pi = \left(rV - rS \frac{\partial V}{\partial S} \right) dt.$$

Finally, we can set our two expressions for $d\Pi$ equal to each other to obtain the famous Black-Scholes equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

2.4 Solving the Black-Scholes equation

In certain cases, such as in the case of a European option, setting V to be equal to a call/put option yields an exact value from the Black-Scholes equation. In other cases, the partial differential equation can be solved using standard methods such as a version of finite differences.

3 Jim Simons: mathematician, professor, code breaker, and “Quant King”

The recent passing of Jim Simons highlights how the Black-Scholes equation and other mathematical models revolutionized trading in the 1980s. After accomplished careers in both academia and government, in 1978, Simons founded an investment firm that would eventually become the hedge fund Renaissance Technologies, which ran one of the most profitable funds of all time, the Medallion Fund. The historic level of returns of this fund effectively disproved the efficient market hypothesis, which states that stock prices reflect all information, and therefore stocks trade at their fair market value. Over a period of 30 years, from 1988 to 2018, the Medallion Fund earned its investors \$100 billion with annual returns of 66 percent (before fees). Simons’ landmark success eclipses that of his contemporaries such as Warren Buffett, George Soros, Peter Lynch, Steve Cohen, and Ray Dalio.

Following in the footsteps of mathematicians from the 19th century through the late 20th century, such as Louis Bachelier, Ed Thorpe, Fischer Black, Myron Sholes, Robert C. Merton, and Leonard Baum, Jim Simons sought to improve and refine trading models by detecting short-term

patterns in option price discovery. He primarily employed mathematicians and scientists (rather than traders) who pioneered the use of algorithmic data analysis to exploit statistical anomalies in the stock market. Quantitative analysis has since spread throughout Wall Street, using machine learning to frequently beat the market.

4 Conclusion

Over the span of the past 50 years, the Black-Scholes equation has contributed to four multi-trillion dollar industries: exchange traded options, over-the-counter derivative securities, credit default swaps, and securitized debt. It fundamentally transformed investors' approach to risk. Some of its limitations have spawned alternative option pricing models, such as the Binomial model and the Monte Carlo simulation model. But the Nobel-prize-winning Black-Scholes Equation's robust mathematical framework remains one of the most impactful applications in the history of finance. By creating entirely new industries, mathematical insight provides unprecedented opportunities to grow investments and improve outcomes and lives all over the world.

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